

CSA0510-DBMS EXPERIMENTS:

1)

1.1)

```
mysql> create table student1( Regno int(3), sname varchar(15), gender char(1), DOB date, Mno int(10), city varchar(20));
Query OK, 0 rows affected, 2 warnings (0.735 sec)

mysql> select * from student1;
Empty set (0.027 sec)

mysql> desc student1;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Regno  | int           | YES  |     | NULL    |       |
| sname  | varchar(15)   | YES  |     | NULL    |       |
| gender | char(1)       | YES  |     | NULL    |       |
| DOB    | date          | YES  |     | NULL    |       |
| Mno    | int           | YES  |     | NULL    |       |
| city   | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.053 sec)
```

1.2)

```
mysql> create table Faculty( Facno varchar(5), Facname varchar(20), gender char(1), DOB DATE, DOJ DATE, mobileno varchar(10));
Query OK, 0 rows affected (0.678 sec)

mysql> desc Faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno  | varchar(5)    | YES  |     | NULL    |       |
| Facname | varchar(20)   | YES  |     | NULL    |       |
| gender | char(1)       | YES  |     | NULL    |       |
| DOB    | date          | YES  |     | NULL    |       |
| DOJ    | date          | YES  |     | NULL    |       |
| mobileno | varchar(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.031 sec)
```

1.3)

```
mysql> create table department( DeptNo varchar(5), DeptName Varchar(20), DeptHead Varchar(10));
Query OK, 0 rows affected (0.635 sec)

mysql> desc department;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo | varchar(5)    | YES  |     | NULL    |       |
| DeptName | varchar(20)  | YES  |     | NULL    |       |
| DeptHead | varchar(10)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.041 sec)
```

1.4)

```
mysql> create table course(CourseNo varchar(5), CourseDesc varchar(15), CourseType char(1), Semno char(1), Hallno char(1), Facno varchar(5));
Query OK, 0 rows affected (0.626 sec)

mysql> desc course;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| CourseNo | varchar(5)  | YES  |     | NULL    |       |
| CourseDesc | varchar(15) | YES  |     | NULL    |       |
| CourseType | char(1)    | YES  |     | NULL    |       |
| Semno    | char(1)    | YES  |     | NULL    |       |
| Hallno   | char(1)    | YES  |     | NULL    |       |
| Facno    | varchar(5) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.036 sec)
```

1.5)

```
mysql> alter table Faculty add dept varchar(10);
Query OK, 0 rows affected (0.928 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | varchar(5) | YES | | NULL | |
| Facname | varchar(20) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| mobileno | varchar(10) | YES | | NULL | |
| dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.028 sec)
```

2)

2.1)

```
mysql> create table student2( Regno int(10), S1Name varchar(20), Age char(2), Mobno int(10), address varchar(20));
Query OK, 0 rows affected, 2 warnings (0.227 sec)

mysql> alter table student2 ADD PRIMARY KEY(Regno);
Query OK, 0 rows affected (0.942 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc student2;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Regno | int | NO | PRI | NULL | |
| S1Name | varchar(20) | YES | | NULL | |
| Age | char(2) | YES | | NULL | |
| Mobno | int | YES | | NULL | |
| address | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.029 sec)
```

2.2)

```
mysql> alter table faculty add primary key(facno);
Query OK, 0 rows affected (0.948 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> alter table faculty add check(gender 'M' or 'F');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''M' or 'F')' at line 1
mysql> alter table faculty add check(gender='M' or 'F');
Query OK, 0 rows affected (1.057 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Facno | varchar(5) | NO | PRI | NULL | |
| Facname | varchar(20) | YES | | NULL | |
| gender | char(1) | YES | | NULL | |
| DOB | date | YES | | NULL | |
| DOJ | date | YES | | NULL | |
| mobileno | varchar(10) | YES | | NULL | |
| dept | varchar(10) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.029 sec)
```

2.3)

```
mysql> ALTER TABLE department add primary key(deptno);
Query OK, 0 rows affected (1.248 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc department;
```

Field	Type	Null	Key	Default	Extra
DeptNo	varchar(5)	NO	PRI	NULL	
DeptName	varchar(20)	YES		NULL	
DeptHead	varchar(10)	YES		NULL	

3 rows in set (0.048 sec)

2.4)

```
mysql> alter table course add primary key (courseno);
Query OK, 0 rows affected (0.931 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc course;
```

Field	Type	Null	Key	Default	Extra
CourseNo	varchar(5)	NO	PRI	NULL	
CourseDesc	varchar(15)	YES		NULL	
CourseType	char(1)	YES		NULL	
Semno	char(1)	YES		NULL	
Hallno	char(1)	YES		NULL	
Facno	varchar(5)	YES		NULL	

6 rows in set (0.035 sec)

2.5)

```
mysql> alter table student2 add unique key(s1name);
Query OK, 0 rows affected (0.629 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student2;
```

Field	Type	Null	Key	Default	Extra
Regno	int	NO	PRI	NULL	
S1Name	varchar(20)	YES	UNI	NULL	
Age	char(2)	YES		NULL	
Mobno	int	YES		NULL	
address	varchar(20)	YES		NULL	

5 rows in set (0.030 sec)

2.6)

```
C Syntax to use near address int not null at line 1
mysql> alter table student1 modify address int not null;
Query OK, 0 rows affected (1.021 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student1;
```

Field	Type	Null	Key	Default	Extra
Regno	int	YES		NULL	
sname	varchar(15)	YES		NULL	
gender	char(1)	YES		NULL	
DOB	date	YES		NULL	
Mno	int	YES		NULL	
city	varchar(20)	YES		NULL	
address	int	NO		NULL	

7 rows in set (0.028 sec)

3)

```
mysql> INSERT INTO student1 VALUES("1","Raj","M","2004-3-2",1234567890,"Chennai"),("2","Shru","F","2005-4-3",987654321,"Chennai"),("3","Sam","M","2006-5-4",246813579,"Chennai");
Query OK, 3 rows affected (0.464 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from student1;
```

Regno	sname	gender	DOB	Mno	city
1	Raj	M	2004-03-02	1234567890	Chennai
2	Shru	F	2005-04-03	987654321	Chennai
3	Sam	M	2006-05-04	246813579	Chennai

3 rows in set (0.012 sec)

4)

4.1)

```
mysql> update student1 set sname="Ram" where Regno=1;
Query OK, 1 row affected (0.468 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from student1;
```

Regno	sname	gender	DOB	Mno	city
1	Ram	M	2004-03-02	1234567890	Chennai
2	Shru	F	2005-04-03	987654321	Chennai
3	Sam	M	2006-05-04	246813579	Chennai

3 rows in set (0.009 sec)

```
mysql> select * from faculty;
```

Facno	Facname	gender	DOB	DOJ	mobilen	dept	resigned
123	YAHOO	m	2001-02-03	2005-06-07	12345678	CS	n
135	FireFox	M	2004-05-06	2008-09-10	192837465	EEE	y
456	GOOGLE	F	2002-03-04	2006-07-08	12345678	ECE	n
789	BING	F	2003-04-05	2007-08-09	987654321	CSBS	n

```
4 rows in set (0.013 sec)
```

```
mysql> delete from faculty where resigned='y';
```

```
Query OK, 1 row affected (0.084 sec)
```

```
mysql> select * from faculty;
```

Facno	Facname	gender	DOB	DOJ	mobilen	dept	resigned
123	YAHOO	m	2001-02-03	2005-06-07	12345678	CS	n
456	GOOGLE	F	2002-03-04	2006-07-08	12345678	ECE	n
789	BING	F	2003-04-05	2007-08-09	987654321	CSBS	n

```
3 rows in set (0.455 sec)
```

```
mysql> select * from faculty;
```

Facno	Facname	gender	DOB	DOJ	mobileno	dept	resigned	age
123	YAHOO	m	2001-02-03	2005-06-07	12345678	CS	n	25
456	GOOGLE	F	2002-03-04	2006-07-08	12345678	ECE	n	30
789	BING	F	2003-04-05	2007-08-09	987654321	CSBS	n	20
987	HOOT	M	2005-06-07	2006-07-08	415277482	CSE	n	65

```
4 rows in set (0.405 sec)
```

```
mysql> delete from faculty where age>=65;
```

Query OK, 1 row affected (0.471 sec)

```
mysql> select * from faculty;
```

Facno	Facname	gender	DOB	DOJ	mobileno	dept	resigned	age
123	YAHOO	m	2001-02-03	2005-06-07	12345678	CS	n	25
456	GOOGLE	F	2002-03-04	2006-07-08	12345678	ECE	n	30
789	BING	F	2003-04-05	2007-08-09	987654321	CSBS	n	20

```
3 rows in set (0.011 sec)
```

5)

5.1)

```
mysql> select reg_no,sname,DOB from student;
+-----+-----+-----+
| reg_no | sname   | DOB      |
+-----+-----+-----+
| 1      | apple   | 2000-01-01 |
| 2      | banana  | 2000-02-02 |
| 3      | Chocolate | 2000-03-03 |
+-----+-----+-----+
3 rows in set (0.010 sec)
```

5.2)

```
mysql> select * from student where gender='F';
+-----+-----+-----+-----+-----+-----+
| reg_no | sname   | gender | dob      | mno      | location |
+-----+-----+-----+-----+-----+-----+
| 1      | apple   | F      | 2000-01-01 | 111111111 | Chennai |
| 3      | Chocolate | F      | 2000-03-03 | 333333333 | Chennai |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.011 sec)
```

5.3)

```
mysql> select * from student where courseno='C001';
+-----+-----+-----+-----+-----+-----+-----+
| reg_no | sname   | gender | dob      | mno      | location | courseno |
+-----+-----+-----+-----+-----+-----+-----+
| 1      | apple   | F      | 2000-01-01 | 111111111 | Chennai | C001     |
| 2      | banana  | M      | 2000-02-02 | 222222222 | Chennai | C001     |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.009 sec)
```

5.4)

```
mysql> select * from faculty where DOJ<'2014-11-1';
+-----+-----+-----+-----+
| faculty_code | faculty_name | specialization | DOJ      |
+-----+-----+-----+-----+
| 1            | Dr. Kumar    | Database Systems | 2010-09-08 |
| 2            | Dr. Ravi     | Operating Systems | 2011-10-09 |
| 3            | Dr. Priya    | Computer Networks | 2009-08-07 |
+-----+-----+-----+-----+
3 rows in set (0.010 sec)
```

5.5)

```
mysql> select * from course where hallno!='H001';
```

CourseNo	CourseDesc	CourseType	Semno	Hallno	Facno
CH101	Chemistry	L	2	H006	F004
CS201	Computer Sci	L	2	H002	F001
MA101	Mathematics	L	1	H003	F002
MA201	Math 201	L	2	H004	F002
PH101	Physics	L	1	H005	F003

5 rows in set (0.402 sec)



5.6)

```
mysql> select * from student where sname like '%sh';  
Empty set (0.011 sec)
```

5.7)

```
mysql> select * from student where sname like '%sh%';  
Empty set (0.011 sec)
```

5.8)

```
mysql> select * from student where location like "%SAL%";  
Empty set (0.011 sec)
```

5.9)

```
mysql> select * from student where sname not like "_____";  
+-----+-----+-----+-----+-----+-----+-----+  
| reg_no | sname | gender | dob | mno | location | courseno |  
+-----+-----+-----+-----+-----+-----+-----+  
| 1 | apple | F | 2000-01-01 | 111111111 | Chennai | C001 |  
| 2 | banana | M | 2000-02-02 | 222222222 | Chennai | C001 |  
| 3 | Chocolate | F | 2000-03-03 | 333333333 | Chennai | C003 |  
+-----+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.010 sec)
```

5.10)

```
mysql> select * from student where sname like "%am%";  
Empty set (0.010 sec)
```

6)

6.1.1)

```
mysql> select coursetype from course where coursedesc in ('ComputerScience', 'Mathematics');
+-----+
| coursetype |
+-----+
| M          |
| E          |
+-----+
2 rows in set (0.010 sec)
```

6.1.2)

```
mysql> SELECT CourseNo FROM register WHERE Mark BETWEEN 50 AND 80;
+-----+
| CourseNo |
+-----+
| C001     |
| C002     |
| C001     |
| C002     |
| C002     |
| C003     |
| C002     |
| C004     |
+-----+
8 rows in set (0.410 sec)
```

6.2.1)

```
mysql> SELECT AVG(Mark) FROM register WHERE CourseNo='C002';
+-----+
| AVG(Mark) |
+-----+
| 66.2500   |
+-----+
1 row in set (0.014 sec)
```

6.2.2)

```
mysql> SELECT MAX(Mark),MIN(Mark) FROM register WHERE CourseNo='C002';
+-----+-----+
| MAX(Mark) | MIN(Mark) |
+-----+-----+
| 80        | 50        |
+-----+-----+
1 row in set (0.013 sec)
```


6.2.3)

```
mysql> SELECT CourseNo,MAX(Mark),MIN(Mark),AVG(Mark) FROM register GROUP BY CourseNo;
```

CourseNo	MAX(Mark)	MIN(Mark)	AVG(Mark)
C001	90	55	73.3333
C002	80	50	66.2500
C003	85	60	72.5000
C004	78	78	78.0000

```
4 rows in set (0.012 sec)
```

6.2.4)

```
mysql> SELECT c.CourseDesc,AVG(r.Mark) FROM course c JOIN register r ON c.CourseNo=r.CourseNo GROUP BY c.CourseNo,c.CourseDesc;
```

CourseDesc	AVG(r.Mark)
ComputerScience	73.3333
Mathematics	66.2500
Statistics	72.5000
Programming	78.0000

```
4 rows in set (0.016 sec)
```

6.2.5)

```
mysql> SELECT SUM(Mark) FROM register;
```

SUM(Mark)
708

```
1 row in set (0.011 sec)
```

6.2.6)

```
mysql> SELECT CourseNo,COUNT(StudentNo) FROM register GROUP BY CourseNo;
```

CourseNo	COUNT(StudentNo)
C001	3
C002	4
C003	2
C004	1

```
4 rows in set (0.011 sec)
```

6.2.7)

```
mysql> SELECT c.CourseDesc,COUNT(r.StudentNo) FROM course c JOIN register r ON c.CourseNo=r.CourseNo GROUP BY c.CourseNo,c.CourseDesc;
```

CourseDesc	COUNT(r.StudentNo)
ComputerScience	3
Mathematics	4
Statistics	2
Programming	1

```
4 rows in set (0.013 sec)
```

6.2.8)

```
mysql> SELECT StudentNo,COUNT(CourseNo) FROM register GROUP BY StudentNo;
```

StudentNo	COUNT(CourseNo)
S001	3
S002	2
S003	2
S004	3

4 rows in set (0.012 sec)

7)

7.1)

```
mysql> SELECT c.CourseDesc,COUNT(r.StudentNo) FROM course c JOIN register r ON c.CourseNo=r.CourseNo GROUP BY c.CourseNo,c.CourseDesc;
```

CourseDesc	COUNT(r.StudentNo)
ComputerScience	3
Mathematics	4
Statistics	2
Programming	1

4 rows in set (0.014 sec)

7.2)

```
mysql> SELECT StudentNo,COUNT(CourseNo) FROM register GROUP BY StudentNo;
```

StudentNo	COUNT(CourseNo)
S001	3
S002	2
S003	2
S004	3

4 rows in set (0.010 sec)

7.3)

```
mysql> SELECT StudentName,StudentNo FROM student ORDER BY StudentNo ASC;
```

StudentName	StudentNo
Arun	S001
Rahul	S002
Kavin	S003
John	S004

4 rows in set (0.010 sec)

7.4)

```
mysql> SELECT * FROM faculty ORDER BY DOJ ASC;
```

faculty_code	faculty_name	specialization	DOJ	hallno
3	Dr. Priya	Computer Networks	2009-08-07	H003
1	Dr. Kumar	Database Systems	2010-09-08	H001
2	Dr. Ravi	Operating Systems	2011-10-09	H002

3 rows in set (0.011 sec)

8)

8.1)

```
mysql> SELECT StudentNo,CourseNo,Mark FROM register WHERE Mark>(SELECT AVG(Mark) FROM register);
```

StudentNo	CourseNo	Mark
S001	C001	75
S001	C003	85
S002	C002	80
S004	C001	90
S004	C004	78

5 rows in set (0.018 sec)

8.2)

```
mysql> SELECT s.StudentName,COUNT(r.CourseNo) FROM student s JOIN register r ON s.StudentNo=r.StudentNo GROUP BY s.StudentNo,s.Studen  
tName HAVING COUNT(r.CourseNo)>1;
```

StudentName	COUNT(r.CourseNo)
Arun	3
Rahul	2
Kavin	2
John	3

4 rows in set (0.016 sec)

8.3)

```
mysql> SELECT faculty_name,DOJ FROM faculty ORDER BY DOJ DESC LIMIT 1;
```

faculty_name	DOJ
Dr. Ravi	2011-10-09

1 row in set (0.010 sec)

8.4)

```
mysql> SELECT CourseNo,Mark FROM register r WHERE Mark>(SELECT AVG(Mark) FROM register r2 WHERE r2.CourseNo=r.CourseNo);
```

CourseNo	Mark
C001	75
C003	85
C002	80
C002	70
C001	90

5 rows in set (0.014 sec)

9)

9.1)

```
mysql> SELECT faculty.faculty_code, faculty.faculty_name, department.DeptNo, department.DeptName FROM faculty, department WHERE faculty.deptno = department.DeptNo;
```

faculty_code	faculty_name	DeptNo	DeptName
1	Dr. Kumar	D01	CSE
2	Dr. Ravi	D02	ECE
3	Dr. Priya	D03	IT

3 rows in set (0.012 sec)

9.2)

```
mysql> SELECT student.StudentName, register.CourseNo FROM student LEFT OUTER JOIN register ON student.StudentNo = register.StudentNo WHERE register.CourseNo IS NULL;
```

StudentName	CourseNo
Meena	NULL

1 row in set (0.012 sec)

9.3)

```
mysql> SELECT course.CourseNo, course.CourseDesc FROM course, faculty, department WHERE course.HallNo = faculty.hallno AND faculty.deptno = department.DeptNo AND department.DeptName = 'Sales' AND faculty.faculty_name = 'Ratnam';
```

CourseNo	CourseDesc
C005	SalesTechniques

1 row in set (0.013 sec)

10)

```
mysql> SELECT * FROM v1;
```

employee_id	salary
10	150000
11	120000
12	160000
13	90000

4 rows in set (0.013 sec)

EXPERIMENT-11:-

JOINS

```
mysql> CREATE TABLE Employees (  
->     emp_id INT PRIMARY KEY,  
->     emp_name VARCHAR(50) NOT NULL,  
->     salary DECIMAL(10,2),  
->     dept_id INT,  
->     FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)  
-> );  
Query OK, 0 rows affected (0.631 sec)
```

```
mysql>  
mysql> -- 4. Create the Projects table  
Query OK, 0 rows affected (0.006 sec)
```

```
mysql> CREATE TABLE Projects (  
->     project_id INT PRIMARY KEY,  
->     project_name VARCHAR(50) NOT NULL,  
->     emp_id INT,  
->     FOREIGN KEY (emp_id) REFERENCES Employees(emp_id)  
-> );  
Query OK, 0 rows affected (1.815 sec)
```

```
mysql>  
mysql> -- 5. Insert sample data into Departments  
Query OK, 0 rows affected (0.005 sec)  
  
mysql> INSERT INTO Departments VALUES (1, 'Human Resources');  
Query OK, 1 row affected (0.217 sec)  
  
mysql> INSERT INTO Departments VALUES (2, 'Software Development');  
Query OK, 1 row affected (0.052 sec)  
  
mysql> INSERT INTO Departments VALUES (3, 'Finance');  
Query OK, 1 row affected (0.053 sec)  
  
mysql>  
mysql> -- 6. Insert sample data into Employees  
Query OK, 0 rows affected (0.005 sec)  
  
mysql> INSERT INTO Employees VALUES (101, 'Alice Smith', 55000, 1);  
Query OK, 1 row affected (0.067 sec)  
  
mysql> INSERT INTO Employees VALUES (102, 'Bob Jones', 62000, 2);  
Query OK, 1 row affected (0.049 sec)  
  
mysql> INSERT INTO Employees VALUES (103, 'Charlie Brown', 48000, 2);  
Query OK, 1 row affected (0.054 sec)  
  
mysql> INSERT INTO Employees VALUES (104, 'Diana Prince', 71000, 3);  
Query OK, 1 row affected (0.055 sec)
```

```
mysql> -- Left Outer Join Query  
Query OK, 0 rows affected (0.004 sec)  
  
mysql> SELECT Employees.emp_name, Projects.project_name  
-> FROM Employees  
-> LEFT JOIN Projects ON Employees.emp_id = Projects.emp_id;  
+-----+-----+  
| emp_name | project_name |  
+-----+-----+  
| Alice Smith | Alpha Portal |  
| Bob Jones | Beta Cloud |  
| Charlie Brown | NULL |  
| Diana Prince | Gamma Analytics |  
+-----+-----+  
4 rows in set (0.016 sec)
```

EXPERIMENT-12

VIEWS AND INDEXES

```
mysql> SHOW INDEX FROM Employees;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null |
| Index_type | Comment | Index_comment | Visible | Expression | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| employees | 0 | PRIMARY | 1 | emp_id | A | 4 | NULL | NULL | |
| BTREE | | | YES | NULL | | | | | |
| employees | 1 | dept_id | 1 | dept_id | A | 3 | NULL | NULL | YES |
| BTREE | | | YES | NULL | | | | | |
| employees | 1 | idx_emp_name | 1 | emp_name | A | 4 | NULL | NULL | |
| BTREE | | | YES | NULL | | | | | |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.720 sec)
```

```
mysql> -- 2. Query the View to verify its creation and output
Query OK, 0 rows affected (0.005 sec)
```

```
mysql> SELECT * FROM high_earners;
ERROR 1146 (42S02): Table 'nig.high_earners' doesn't exist
mysql>
```

```
mysql> -- 3. Create an Index on the employee name column for faster searching
Query OK, 0 rows affected (0.006 sec)
```

```
mysql> CREATE INDEX idx_emp_name ON Employees(emp_name);
Query OK, 0 rows affected (0.989 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> -- 4. Verify the index creation (MySQL syntax)
Query OK, 0 rows affected (0.005 sec)
```

EXPERIMENT-13

INTRODUCTION TO PROCEDURAL SQL / CONTROL STRUCTURES

```
mysql>
-- Conditional control structure (IF-THEN-ELSE)
-- IF v_sal < 60000 THEN
-- UPDATE Employees SET salary = salary + 5000 WHERE emp_id = p_emp_id;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'IF v_sal < 60000 THEN
UPDATE Employees SET salary = salary + 5000 WHERE ' at line 2
mysql> SELECT CONCAT('Salary updated for employee ID: ', p_emp_id) AS Result;
ERROR 1054 (42S22): Unknown column 'p_emp_id' in 'field list'
mysql> ELSE
-- SELECT CONCAT('Salary is already above or equal to threshold for employee ID: ', p_emp_id) AS Result;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'ELSE
SELECT CONCAT('Salary is already above or equal to threshold for em' at line 1
mysql> END IF;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'END IF' at line 1
mysql> END //
--
-- DELIMITER ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'END //
```

```

DELIMITER' at line 1
mysql>
mysql> -- Execute the procedure
Query OK, 0 rows affected (0.003 sec)

mysql> CALL CheckAndRaiseSalary(101);
ERROR 1305 (42000): PROCEDURE nig.CheckAndRaiseSalary does not exist
mysql>
mysql> -- Verify the update
Query OK, 0 rows affected (0.005 sec)

mysql> SELECT emp_id, emp_name, salary FROM Employees WHERE emp_id = 101;
+-----+-----+-----+
| emp_id | emp_name | salary |
+-----+-----+-----+
| 101 | Alice Smith | 55000.00 |
+-----+-----+-----+
1 row in set (0.007 sec)

```

EXPERIMENT-14

CURSORS AND LOOPS

```

mysql> CALL ProcessEmployeeCursor();
ERROR 1305 (42000): PROCEDURE nig.ProcessEmployeeCursor does not exist
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE ProcessEmployeeCursor()
-> BEGIN
->     DECLARE done INT DEFAULT FALSE;
->     DECLARE v_name VARCHAR(50);
->     DECLARE v_sal DECIMAL(10,2);
->
->     DECLARE emp_cursor CURSOR FOR SELECT emp_name, salary FROM Employees;
->     DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = TRUE;
->
->     OPEN emp_cursor;
->
->     read_loop: LOOP
->         FETCH emp_cursor INTO v_name, v_sal;
->         IF done THEN
->             LEAVE read_loop;
->         END IF;
->         SELECT CONCAT('Name: ', v_name, ' | Salary: ', v_sal) AS Employee_Info;
->     END LOOP;

```

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> -- Now execute it successfully
Query OK, 0 rows affected (0.002 sec)

mysql> CALL ProcessEmployeeCursor();
+-----+
| Employee_Info |
+-----+
| Name: Alice Smith | Salary: 55000.00 |
+-----+
1 row in set (0.334 sec)

+-----+
| Employee_Info |
+-----+
| Name: Bob Jones | Salary: 62000.00 |
+-----+
1 row in set (0.475 sec)

```

EXPERIMENT-15

STORED PROCEDURES

```
--> DELIMITER //
```

```
--> CREATE PROCEDURE UpdateEmployeeSalary(  
-->     IN p_emp_id INT,  
-->     IN p_increment DECIMAL(10,2)  
--> )  
--> BEGIN  
-->     UPDATE Employees  
-->     SET salary = salary + p_increment  
-->     WHERE emp_id = p_emp_id;  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version  
for the right syntax to use near '-----' at  
line 1  
mysql>  
-->     SELECT CONCAT('Successfully updated salary for Employee ID: ', p_emp_id) AS Status;  
ERROR 1054 (42S22): Unknown column 'p_emp_id' in 'field list'  
mysql> END //
```

```
--> DELIMITER ;  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version  
for the right syntax to use near 'END //
```

```
DELIMITER' at line 1  
mysql>  
mysql> -- Execute Experiment 15  
Query OK, 0 rows affected (0.002 sec)  
  
mysql> CALL UpdateEmployeeSalary(102, 3000);  
ERROR 1305 (42000): PROCEDURE nlg.UpdateEmployeeSalary does not exist  
mysql> SELECT * FROM Employees WHERE emp_id = 102;  
+-----+-----+-----+-----+  
| emp_id | emp_name | salary | dept_id |  
+-----+-----+-----+-----+  
|    102 | Bob Jones | 62000.00 |      2 |  
+-----+-----+-----+-----+  
1 row in set (0.005 sec)
```

EXPERIMENT-16

STORED FUNCTIONS

```
mysql>  
mysql> DELIMITER //
```

```
mysql>  
mysql> CREATE FUNCTION GetAnnualSalary(p_emp_id INT)  
--> RETURNS DECIMAL(10,2)  
--> DETERMINISTIC  
--> BEGIN  
-->     DECLARE v_annual_sal DECIMAL(10,2);  
-->  
-->     SELECT salary * 12 INTO v_annual_sal  
-->     FROM Employees  
-->     WHERE emp_id = p_emp_id;  
-->  
-->     RETURN v_annual_sal;  
--> END //
```

```
Query OK, 0 rows affected (0.160 sec)
```

```
mysql> SELECT emp_id, emp_name, GetAnnualSalary(101) AS annual_salary  
--> FROM Employees  
--> WHERE emp_id = 101;  
+-----+-----+-----+  
| emp_id | emp_name | annual_salary |  
+-----+-----+-----+  
|    101 | Alice Smith | 660000.00 |  
+-----+-----+-----+  
1 row in set (0.115 sec)
```


EXPERIMENT-17

EXCEPTION HANDLING

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE GetEmployeeName(IN p_emp_id INT)
```

```
  -> BEGIN
```

```
  ->   DECLARE v_name VARCHAR(50);
```

```
  ->   DECLARE EXIT HANDLER FOR NOT FOUND
```

```
  ->   BEGIN
```

```
  ->     SELECT 'Error: Employee ID does not exist.' AS Error_Message;
```

```
  ->   END;
```

```
  ->
```

```
  ->   SELECT emp_name INTO v_name
```

```
  ->   FROM Employees
```

```
  ->   WHERE emp_id = p_emp_id;
```

```
  ->
```

```
  ->   SELECT CONCAT('Employee Name: ', v_name) AS Result;
```

```
  -> END //
```

```
Query OK, 0 rows affected (0.120 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> -- Test the procedure with a valid ID and an invalid ID
```

```
Query OK, 0 rows affected (0.006 sec)
```

```
mysql> CALL GetEmployeeName(101);
```

Result
Employee Name: Alice Smith

```
1 row in set (0.017 sec)
```

```
Query OK, 0 rows affected (0.081 sec)
```

EXPERIMENT-18

DATABASE PACKAGES (SIMULATED VIA STORED PROCEDURES IN MYSQL)

```
mysql> END //
```

```
  ->
```

```
  -> CREATE PROCEDURE EmpPkg_GetCount()
```

```
  -> BEGIN
```

```
  ->   SELECT COUNT(*) AS Total_Employees FROM Employees;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'END //'
```

```
CREATE PROCEDURE EmpPkg_GetCount()
```

```
BEGIN
```

```
  SELECT COUNT(*) AS Total_Em' at line 1
```

```
mysql> END //
```

```
  ->
```

```
  -> DELIMITER ;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'END //'
```

```
DELIMITER' at line 1
```

```
mysql>
```

```
mysql> -- Execute Experiment 18
```

```
Query OK, 0 rows affected (0.004 sec)
```

```
mysql> CALL EmpPkg_AddEmployee(105, 'John Doe', 45000, 1);
```

```
ERROR 1305 (42000): PROCEDURE nlg.EmpPkg_AddEmployee does not exist
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> -- Now execute the procedures
Query OK, 0 rows affected (0.004 sec)

mysql> CALL EmpPkg_AddEmployee(105, 'John Doe', 45000, 1);
+-----+
| Message |
+-----+
| Employee added successfully. |
+-----+
1 row in set (0.092 sec)
```

Query OK, 0 rows affected (0.145 sec)

```
mysql> CALL EmpPkg_GetCount();
+-----+
| Total_Employees |
+-----+
| 5 |
+-----+
1 row in set (0.010 sec)
```

Query OK, 0 rows affected (0.047 sec)

EXPERIMENT-19

DATABASE TRIGGERS

```
DELIMITER' at line 1
mysql>
mysql> -- Execute Experiment 19 (This will trigger an error and block the insert)
Query OK, 0 rows affected (0.002 sec)

mysql> -- INSERT INTO Employees VALUES (106, 'Bad Data', -500, 1);
Query OK, 0 rows affected (0.004 sec)
```

EXPERIMENT 20:

DATABASE SECURITY AND USER MANAGEMENT (DCL)

```
--> mysql> FLUSH PRIVILEGES;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'Query OK, 0 rows affected (0.012 sec)'

mysql> FLUSH PRIVILEGES' at line 1
mysql> Query OK, 0 rows affected (0.005 sec)

-->
--> mysql> REVOKE INSERT ON nig.Employees FROM 'analyst'@'localhost';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'Query OK, 0 rows affected (0.005 sec)'

mysql> REVOKE INSERT ON nig.Employees FRO' at line 1
mysql> Query OK, 0 rows affected (0.007 sec)

-->
--> mysql> SHOW GRANTS FOR 'analyst'@'localhost';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version
for the right syntax to use near 'Query OK, 0 rows affected (0.007 sec)'

mysql> SHOW GRANTS FOR 'analyst'@'localho' at line 1
mysql> +-----+
--> | Grants for analyst@localhost |
--> +-----+
--> | GRANT USAGE ON *.* TO 'analyst'@'localhost' |
--> | GRANT SELECT ON `nig`.`Employees` TO 'analyst'@'lo.. |
--> +-----+
```

